

Test Plan - VWO-49: [VWO] Add Passkey and SSO Login Options to VWO Login Page

Status: **DRAFT - pending human review.** Not approved until a QA owner signs off.

Ticket Snapshot

- **Jira key:** VWO-49
- **Type:** Story
- **Jira priority field:** Medium
- **Description priority:** High
- **Epic:** Authentication Enhancements
- **Components:** Not specified
- **Fix version:** Not specified
- **Labels:** Not specified
- **Linked issues:** None
- **Attachments:** None

1. Scope & Objectives

• In scope:

- Verify the VWO login page displays Sign in using SSO and Sign in with Passkey below Sign in with Google.
- Verify successful and failed SSO login behavior for configured identity providers.
- Verify successful and failed passkey login behavior for registered passkeys, missing passkeys, unsupported devices, and unsupported browsers.
- Verify user redirect to dashboard after successful authentication.
- Verify authentication error messaging, security behavior, accessibility, responsiveness, and regression impact on existing login methods.

• Out of scope:

- Passkey registration flow, because the ticket explicitly says it is covered in a separate story.
- SSO provider configuration, because the ticket explicitly excludes it.
- Backend identity-provider provisioning unless required to execute the login flow.
- New dashboard functionality after login.

• **Objective:**

- Testing means proving that the new SSO and Passkey login options are visible, usable, secure, accessible, compatible with supported environments, and do not regress existing login methods.

2. Gaps & Questions for the author

Requirement Checklist Result

Area	Checklist item	Status	Notes
Functional	Clear user story / goal	PASS	Story explains secure and convenient login via SSO or Passkey.
Functional	Acceptance criteria are testable	REVIEW	Core paths are testable, but wording like proper, fallback, supported, and configured is not concrete.
Functional	Happy path fully described	REVIEW	SSO and Passkey success paths are present, but provider selection, callback, account mapping, and session behavior are not defined.
Functional	Negative / error paths described	REVIEW	Error categories are listed, but exact trigger conditions and expected messages are missing.
Functional	Boundary and empty states	REVIEW	No-passkey state is listed, but first-time user, multiple passkeys, disabled passkey, multiple IdPs, and locked accounts are not covered.
Functional	State transitions / workflow steps	REVIEW	Login to dashboard is described, but redirect/callback/session transitions are not enumerated.
Data & environment	Required test data specified	MISSING	No test users, IdPs, registered passkeys, account states, or device data are named.
Data & environment	Environment / config / feature flags named	MISSING	DoD mentions staging, but no environment URL, flag, tenant, or rollout config is specified.
Data & environment	External dependencies and integrations listed	REVIEW	SSO, OAuth/SAML, and WebAuthn are implied, but actual providers and passkey platforms are missing.
Data & environment	Preconditions / setup stated	REVIEW	Some preconditions are implied, but user state and provider setup are not listed.
Non-functional	Performance / load expectations	MISSING	No login-page or authentication redirect timing expectations are specified.
Non-functional	Security / authorization	REVIEW	Secure protocols and no sensitive exposure are stated, but concrete controls are not listed.
Non-functional	Accessibility expectations	REVIEW	Accessibility is mentioned in test scenarios, but WCAG target and expected keyboard/screen-reader behavior are absent.
Non-functional	Internationalization / localization	MISSING	No language, locale, or translated error-message expectations are specified.
Non-functional	Audit / logging / observability	MISSING	No login event, failure event, or security telemetry expectations are specified.
Cross-cutting	Impact on existing features	MISSING	Regression scope for email/password, Google login, forgot password, and account verification is not specified.
Cross-cutting	Backward compatibility / migration	MISSING	No behavior is defined for existing users without passkeys or SSO.
Cross-cutting	Mobile / responsive / browser matrix	REVIEW	Supported browsers/devices are referenced but not named.

Area	Checklist item	Status	Notes
Cross-cutting	Rollback / feature-flag behavior	MISSING	No rollback or flag behavior is specified.
Clarity	No ambiguous wording	REVIEW	Ambiguous terms include seamlessly, smooth, proper error messages, fallback or guidance, supported browsers/devices, and configured identity providers.
Clarity	Terms defined consistently	REVIEW	Passkey, SSO, supported environment, and configured provider are not explicitly defined.
Clarity	Mockups / designs linked	MISSING	No screenshot, Figma, or visual placement reference is attached.

Open Questions

#	Area	Finding	Question to author
1	Priority	Jira priority is Medium but description says High.	Which priority should QA use for risk and release gating?
2	SSO provider	"All configured identity providers" is not enumerated.	Which IdPs must be tested for this story, such as Okta, Azure AD, Google Workspace, OneLogin, or custom SAML?
3	SSO entry flow	Button behavior is not described.	After clicking Sign in using SSO, does the user enter email/domain first, select an IdP, or get redirected directly?
4	SSO errors	Error message text is not defined.	What exact messages should appear for invalid credentials, canceled IdP login, disabled user, unassigned user, expired session, and provider outage?
5	Account mapping	User provisioning rules are missing.	Should SSO auto-provision users, require an existing VWO user, or block unknown users?
6	Passkey support	Supported browser/device matrix is missing.	Which browsers, OS versions, devices, and authenticators are officially supported for passkey login?
7	Passkey fallback	"Fallback or guidance message" is vague.	What should happen if no passkey exists: show message only, suggest password login, route to registration, or show support guidance?
8	Passkey failures	Failure states are broad.	What should happen for biometric failure, user cancellation, authenticator timeout, revoked passkey, and mismatched account?
9	Existing login regression	Regression scope is not specified.	Should email/password, Google login, forgot password, account verification, and logout be included in release sign-off?
10	Security	Protocol requirements are high level.	Which security checks are mandatory: CSRF state validation, secure cookies, no token in URL, no PII in logs, replay protection, rate limits?
11	Accessibility	Accessibility target is missing.	Should this meet WCAG 2.2 AA, and are there required labels, focus order, and keyboard-only flows?
12	Performance	No timing target is defined.	What are acceptable page-load, IdP redirect, passkey prompt, and dashboard redirect timings?
13	Observability	Audit logging is absent.	Should successful and failed SSO/passkey attempts be logged in audit/security telemetry?
14	Feature control	Rollback behavior is absent.	Is this behind a feature flag, tenant allowlist, or phased rollout control?

#	Area	Finding	Question to author
15	Design	No mockup is linked.	Is there an approved design for button copy, order, spacing, icons, responsive placement, and disabled/unsupported states?

3. Test Scenarios

ID	Priority	Type	Scenario	Maps to
TS-01	P0	Positive	Open the login page and verify Sign in using SSO and Sign in with Passkey are visible below Sign in with Google.	AC1
TS-02	P0	Regression	Verify existing email/password login still appears and authenticates successfully.	Gap 9
TS-03	P0	Regression	Verify existing Google login still appears above the new buttons and authenticates successfully.	AC1, Gap 9
TS-04	P1	UI	Verify button copy, order, spacing, icons if applicable, hover/focus states, and disabled states match approved design.	AC1, Gap 15
TS-05	P1	Responsive	Verify button layout on desktop, tablet, and mobile breakpoints without overlap or clipping.	AC5, Gap 6
TS-06	P0	Positive	Click Sign in using SSO, complete valid IdP authentication, and verify redirect to the dashboard.	AC2
TS-07	P0	Negative	Attempt SSO with invalid credentials and verify the user remains unauthenticated with a clear error message.	AC2, AC4
TS-08	P0	Negative	Cancel SSO authentication at the IdP and verify the login page handles the return safely.	AC4, Gap 4
TS-09	P0	Negative	Attempt SSO with a valid IdP user who is not authorized for the VWO account and verify access is blocked.	Gap 5
TS-10	P0	Security	Verify SSO callback validates state/nonce and rejects tampered or replayed callback requests.	AC6, Gap 10
TS-11	P1	Integration	Verify each configured IdP follows the expected login and error behavior.	AC5, Gap 2
TS-12	P1	Boundary	Verify SSO behavior for a user mapped to multiple VWO accounts or workspaces.	Gap 5
TS-13	P0	Positive	Click Sign in with Passkey, authenticate with a registered passkey, and verify redirect to dashboard.	AC3
TS-14	P0	Negative	Attempt passkey login when no passkey is registered and verify the expected guidance/fallback message.	AC3, AC4, Gap 7
TS-15	P0	Negative	Fail passkey authentication through biometric/PIN failure and verify the user remains unauthenticated.	AC4, Gap 8
TS-16	P1	Negative	Cancel the browser passkey prompt and verify the login page recovers without stale loading state.	AC4, Gap 8
TS-17	P1	Boundary	Let the passkey prompt time out and verify the expected timeout handling and retry option.	AC4, Gap 8
TS-18	P1	Boundary	Verify behavior when multiple registered passkeys are available for the same user.	Gap 8
TS-19	P0	Compatibility	Verify passkey login on supported browsers/devices from the approved matrix.	AC5, Gap 6
TS-20	P0	Compatibility	Verify unsupported browsers/devices show a clear unsupported/guidance message and do not start a broken passkey flow.	AC4, AC5

ID	Priority	Type	Scenario	Maps to
TS-21	P0	Security	Verify SSO and Passkey login do not expose credentials, assertions, access tokens, or sensitive identifiers in UI, URLs, console logs, or network logs.	AC6, Gap 10
TS-22	P0	Security	Verify authenticated sessions use secure cookies/session storage and cannot be created after failed SSO/passkey attempts.	AC6
TS-23	P1	Accessibility	Verify both new buttons are keyboard reachable, have visible focus, have accessible names, and work with screen-reader navigation.	Gap 11
TS-24	P1	Accessibility	Verify error and guidance messages are announced to assistive technologies and are associated with the triggering control.	AC4, Gap 11
TS-25	P1	Performance	Measure login page load, SSO redirect round trip, passkey prompt start, and dashboard redirect against agreed targets.	Gap 12
TS-26	P2	Localization	Verify new button labels and error/guidance messages are localizable or remain acceptable in supported locales.	Gap 14
TS-27	P1	Observability	Verify successful and failed SSO/passkey login attempts emit required security/audit events if logging is in scope.	Gap 13
TS-28	P1	Rollback	If feature flagged, disable the flag and verify the new buttons disappear while existing login methods continue working.	Gap 14
TS-29	P1	Error handling	Simulate IdP outage/network failure and verify a clear recoverable error, retry path, and no partial login.	AC4, Gap 4
TS-30	P2	Browser history	After failed and successful SSO/passkey attempts, verify back/refresh behavior does not expose stale auth state.	AC6

4. Test Data & Environment

• Data:

- User with existing VWO account and valid email/password credentials.
- User with valid Google login, if Google auth is available in the environment.
- User with valid SSO access for each configured IdP.
- User with valid IdP credentials but no VWO account or no account access.
- User with registered passkey.
- User with no registered passkey.
- User with multiple registered passkeys, if supported.
- Locked, disabled, or deactivated user, if supported by current authentication rules.

• Environment / flags:

- Staging or QA environment with the new login options deployed.

- VWO tenant/account configured for SSO login.
- Passkey/WebAuthn enabled over HTTPS.
- Feature flag or tenant allowlist details, if applicable.
- Browser/device matrix approved by product and engineering.
- **Roles / permissions:**
- Standard active user.
- Admin user if SSO/passkey behavior varies by role.
- Deactivated or access-revoked user.
- New/unprovisioned SSO user if account mapping is in scope.

5. Risks & Assumptions

- **Assumptions made:**
- Sign in with Google already exists and must remain above the new SSO and Passkey buttons.
- A successful login by any supported method lands the user on the same default dashboard.
- SSO provider setup and passkey registration are already available in the test environment.
- The new options are enabled for the target test tenant.
- Security validation is required because the story changes authentication entry points.
- **Risks:**
- Priority mismatch can cause under-testing if the team treats this as Medium while the description says High.
- Passkey behavior varies heavily by OS, browser, authenticator, and device state.
- SSO behavior can differ across IdPs and account provisioning models.
- Missing design references can cause UI placement disputes late in QA.
- Weak error handling can lock users out or leak sensitive authentication details.
- Regression in existing login methods would block all users, even if new methods work.

6. Entry / Exit criteria

- **Entry:**
- VWO-49 is deployed to a QA/staging environment.

- Test tenant has SSO enabled and at least one configured IdP.
- Test tenant has passkey login enabled and at least one user with a registered passkey.
- Existing login methods are available for regression testing.
- Product/engineering confirms supported browser/device matrix.
- Expected error/guidance messages are provided or approved during review.
- **Exit:**
- All P0 scenarios are executed and pass, or have approved release-risk exceptions.
- All P1 scenarios are executed or explicitly deferred by QA/product.
- No Critical or High defects remain open for authentication success, authentication failure, security, or existing login regression.
- Product/engineering answers open questions that affect sign-off.
- QA owner reviews and approves the draft test plan.

HUMAN REVIEW GATE

- **I assumed:**
- SSO and passkey setup already exists in the target environment.
- Existing email/password and Google login are regression-critical.
- The new buttons are part of the same login page and not a separate route.
- Dashboard redirect is the expected success destination for both new methods.
- **I could not confirm:**
- Exact IdPs to test.
- Exact supported browser/device matrix for passkeys.
- Exact error and guidance message text.
- Whether the feature is behind a flag or tenant rollout.
- Whether audit/security telemetry is required.
- Whether WCAG 2.2 AA is the accessibility target.
- **Open questions blocking sign-off:**
- Resolve the priority mismatch: Medium in Jira field vs High in description.
- Provide IdP list and SSO entry flow behavior.

- Provide supported passkey browser/device matrix.
- Provide expected error/guidance copy.
- Confirm whether existing login methods are part of release sign-off.
- Confirm security, performance, accessibility, logging, and rollback requirements.
 - **Reviewer action:**
- Approve or edit this draft before test cases or automation are written.